

SHETH MVLU COLLEGE

SUBJECT:- WSN

AIM:Energy-Aware MAC Protocol for WSN

Implement duty-cycling MAC protocol and evaluate energy conservation in sensor networks.

CODE:-

```
set ns [new Simulator]
```

```
set tr [open energy.tr w]
```

```
$ns trace-all $tr
```

```
set nf [open energy.nam w]
```

```
$ns namtrace-all-wireless $nf 500 500
```

```
set topo [new Topography]
```

```
$topo load_flatgrid 500 500
```

```
create-god 5
```

```
set chan [new Channel/WirelessChannel]
```

```
$ns node-config \
```

```
-adhocRouting DSDV \
```

```
-llType LL \
```

```
-macType Mac/802_11 \
```

```
-ifqType Queue/DropTail/PriQueue \
```

```
-ifqLen 50 \
```

```
-antType Antenna/OmniAntenna \
```

```
-propType Propagation/TwoRayGround \
```

```
-phyType Phy/WirelessPhy \
```

```
-channel $chan \
```

```
-topoInstance $topo \
```

```
-agentTrace ON \
```

```
-routerTrace ON \
```

```
-macTrace ON \
```

```
-movementTrace ON \
```

```
-energyModel EnergyModel \
```

```
-initialEnergy 10 \
```

```
-txPower 2.0 \
```

```
-rxPower 1.0 \
```

```
-idlePower 0.5
```

```
for {set i 0} {$i < 5} {incr i} {
```

```
    set node_($i)[ns node]
```

```
}
```

SHETH MVLU COLLEGE

SUBJECT:- WSN

```
$node_(0) set X_ 50  
$node_(0) set Y_ 250  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 150  
$node_(1) set Y_ 250  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 350  
$node_(3) set Y_ 250  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 450  
$node_(4) set Y_ 250  
$node_(4) set Z_ 0
```

```
for {set i 0} {$i < 5} {incr i} {  
    $ns initial_node_pos node_$i 20  
}
```

```
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

```
set null0 [new Agent/Null]  
$ns attach-agent $node_(4) $null0
```

```
$ns connect $udp0 $null0
```

```
set cbr0 [new Application/Traffic/CBR]  
$cbr0 set packetSize_ 1000  
$cbr0 set interval_ 0.02  
$cbr0 attach-agent $udp0
```

```
nsat1.0 "cbr0 start"  
nsat20.0 "cbr0 stop"
```

```
proc finish {} {  
    global ns tr nf node_
```

SHETH MVLU COLLEGE

SUBJECT:- WSN

```
puts "\n==== ENERGY REPORT ====="
```

```
for {set i 0} {$i < 5} {incr i} {  
    puts "Node i Remaining Energy = [node_($i) energy] J"  
}
```

```
$ns flush-trace
```

```
close $tr
```

```
close $nf
```

```
exec nam energy.nam &
```

```
exit 0
```

```
}
```

```
$ns at 25.0 "finish"
```

```
puts "Simulation Started..."
```

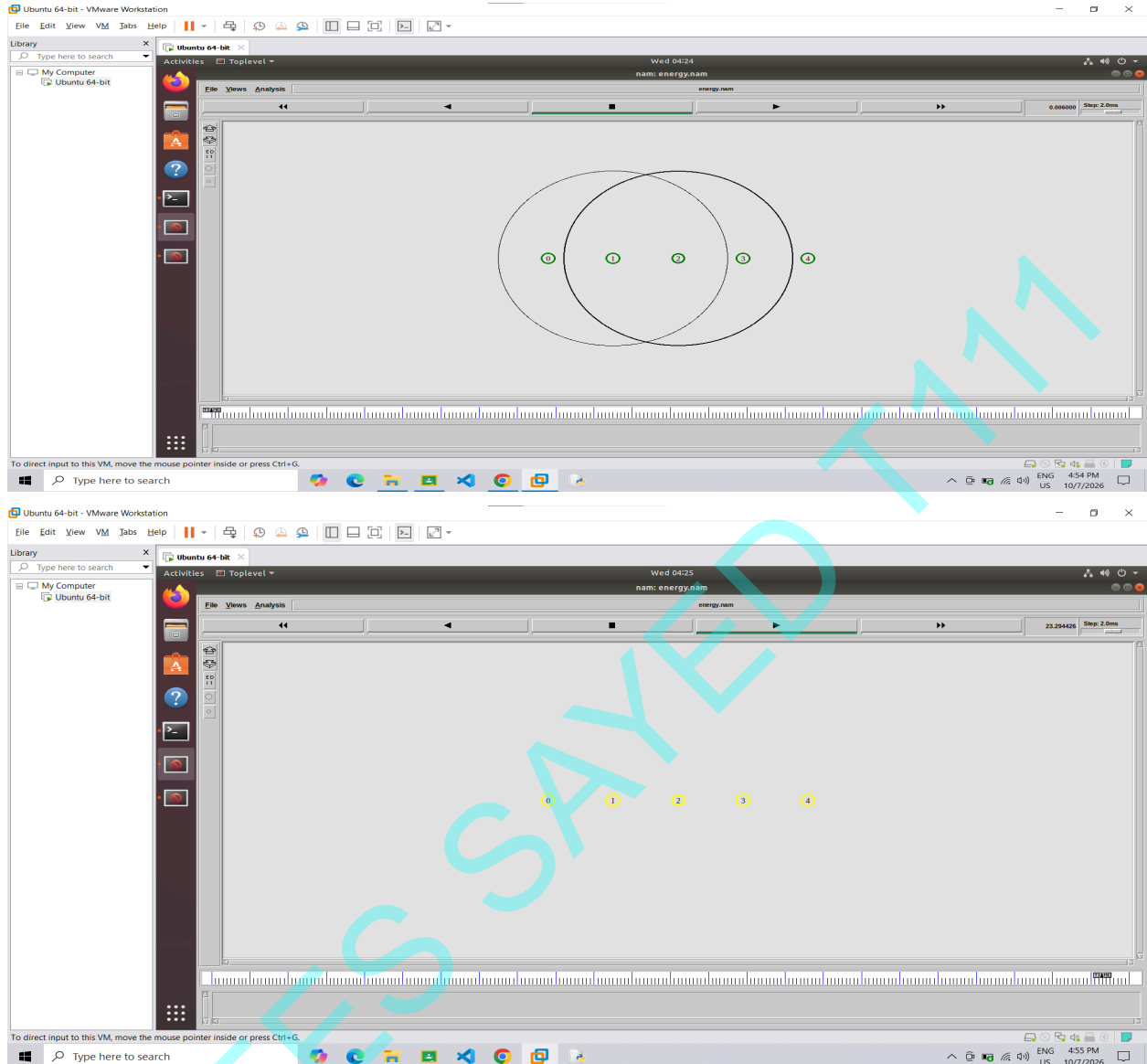
```
$ns run
```

OUTPUT:-

```
anees@ubuntu:~/MANET_Practicals$ cd ~/MANET_Practicals  
anees@ubuntu:~/MANET_Practicals$ nano energy_nac.tcl  
anees@ubuntu:~/MANET_Practicals$ ns energy_nac.tcl  
When configured, ns found the right version of tclsh in /usr/bin/tclsh0.6  
but it doesn't seem to be there anymore, so ns will fall back on running the first tclsh in your path. The wrong version of tclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.  
num_nodes is set 5  
INITIALIZE THE LIST xListHead  
Simulation Started...  
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_  
highestAntennaZ_ = 1.5, distCST_ = 550.0  
SORTING LISTS ...DONE!  
  
==== ENERGY REPORT =====  
Node 0 Remaining Energy = 0.000000 J  
Node 1 Remaining Energy = 0.000000 J  
Node 2 Remaining Energy = 0.000000 J  
Node 3 Remaining Energy = 0.000000 J  
Node 4 Remaining Energy = 0.000000 J  
anees@ubuntu:~/MANET_Practicals$
```

SHETH MVLU COLLEGE

SUBJECT:- WSN



Observation

- Five sensor nodes were deployed in a linear topology.
- Continuous CBR traffic was generated from Node 0 to Node 4.
- Energy decreased during packet transmission and reception.
- Nodes involved in forwarding packets consumed more energy than idle nodes.
- Remaining energy values were displayed after simulation completion.

Result

The energy-aware WSN simulation was successfully executed. Energy consumption was observed for all sensor nodes, and the remaining energy levels were measured after data transmission.